UNIT 1: SOLUTIONS

Questions 1–72

1. [M] Define a solution and distinguish between a saturated, unsaturated and supersaturated solution.

2. [M] Explain why a solution of a non-volatile solute has a lower vapour pressure than the pure solvent.

3. [M] State Raoult's law for a solution containing two volatile components.

4. [H] Derive the relation between relative lowering of vapour pressure and mole fraction of a non-volatile solute.

5. [M] What is an ideal solution? Give two conditions for ideal behaviour.

6. [H] Explain positive and negative deviations from Raoult's law with suitable examples.

7. [M] Why do some liquid pairs show positive deviation from Raoult's law?

8. [H] A liquid mixture shows negative deviation from Raoult's law. Predict the signs of ΔHmix and ΔVmix and justify.

9. [M] Define molarity, molality, mole fraction and mass percentage.

10. [H] Why is molality preferred over molarity when studying colligative properties?

11. [M] Calculate the molarity of a solution containing 5.85 g NaCl in 500 mL of solution.

12. [H] A solution is prepared by dissolving 18 g glucose in 180 g water. Calculate the mole fractions of glucose and water.

13. [M] Define solubility and state the effect of temperature on the solubility of most solids in liquids.

14. [M] State Henry's law and write its mathematical expression.

15. [H] Explain why the solubility of a gas in a liquid generally decreases when temperature increases.

16. [M] Why are soft drinks bottled under high pressure?

17. [H] At a fixed temperature, explain how the mole fraction of a dissolved gas depends on its partial pressure.

18. [M] What are solid solutions? Give one example of a substitutional solid solution.

19. [M] Define colligative properties and list the four principal colligative properties.

20.[H] Why do colligative properties depend on the number of solute particles rather than their chemical identity?

21.[M] Define relative lowering of vapour pressure.

22.[M] Define elevation in boiling point and write its relation to molality.

23.[M] Define depression in freezing point and write its relation to molality.

24.[M] What is osmotic pressure?

25.[H] Derive the equation $\pi = CRT$ for a dilute solution and state the assumptions involved.

26.[M] Why is osmotic pressure useful for determining the molar mass of macromolecules?

27.[H] Compare reverse osmosis and osmosis. Mention one practical application of reverse osmosis.

28. [M] What are isotonic, hypertonic and hypotonic solutions?

29.[H] Predict the direction of solvent flow when two solutions of different osmotic pressures are separated by a semipermeable membrane.

30. [M] What is an abnormal molar mass?

31.[M] Define the van't Hoff factor and write its relation with observed and calculated colligative properties.

32.[H] Explain how association of solute particles affects the van't Hoff factor and observed molar mass.

33.[H] Explain how dissociation of a solute affects its van't Hoff factor and observed molar mass.

34.[M] Why does NaCl show a van't Hoff factor greater than one in a dilute aqueous solution?

35.[H] For a solute that dimerises, derive the relation between degree of association and van't Hoff factor.

36. [H] For a solute that dissociates into n particles, derive the relation between degree of dissociation and van't Hoff factor.

37. [M] A solution contains 2 g of a non-volatile solute in 100 g water. What data are needed to calculate its molar mass using freezing-point depression?

38. [H] Calculate the freezing-point depression for a 0.20 mol kg^{-1} aqueous solution of a non-electrolyte. Kf for water is 1.86 K kg mol^{-1}.

39. [H] A 0.1 mol kg^{-1} solution of a solute has $\Delta Tf = 0.279$ K in water. Calculate its van't Hoff factor. $Kf = 1.86$ K kg mol^{-1}.

40. [H] A solution contains 1.0 g of a polymer in 100 mL solution and has an osmotic pressure of 0.0246 atm at 300 K. Estimate its molar mass using $R = 0.0821$ L atm mol^{-1} K^{-1}.

41. [M] Why is the boiling point of a solution containing a non-volatile solute higher than that of the pure solvent?

42. [H] Explain why adding ethylene glycol to water lowers the freezing point of the mixture.

43. [M] State the relationship between elevation in boiling point and depression in freezing point for the same molality.

44. [H] A solution has a higher vapour pressure than predicted by Raoult's law. What can you infer about solute–solvent interactions?

45. [M] Distinguish between ideal and non-ideal solutions.

46. [H] Explain why a mixture of ethanol and acetone may show positive deviation from Raoult's law.

47. [M] What is an azeotrope? Distinguish between minimum-boiling and maximum-boiling azeotropes.

48. [H] Explain why the composition of an azeotrope cannot be separated completely by simple fractional distillation.

49. [M] Why is the vapour pressure of a pure liquid affected by temperature?

50. [H] A solution is prepared by mixing equal masses of two liquids with different molar masses. Explain why their mole fractions will not generally be equal.

UNIT 1 — Additional Practice

51. [M] Calculate the mass percentage of solute in a solution made by dissolving 12 g solute in 108 g solvent.

52. [H] A 2 molal aqueous solution is prepared. Calculate the moles of solute present per 500 g of water.

53. [M] Convert a solution containing 10% (w/w) solute into a description of what the percentage means.

54. [H] A solution contains 5 g urea in 95 g water. Calculate its molality.

55. [M] Explain why the vapour pressure of a solution changes when the mole fraction of a volatile component changes.

56. [H] For an ideal binary solution, write expressions for total vapour pressure and partial vapour pressures.

57. [M] What is the significance of the slope of a vapour-pressure versus mole-fraction plot for an ideal solution?

58. [H] Explain how intermolecular interactions determine whether mixing produces positive or negative deviation.

59. [M] Why does a gas dissolve more readily in a liquid at higher pressure?

60. [H] A gas has Henry's law constant $k_H = 1.0 \times 10^5$ kPa and partial pressure 50 kPa. Calculate its mole fraction in solution using $p = k_H x$.

61. [M] Distinguish between molality and mole fraction.

62. [H] Calculate the boiling-point elevation of a 0.50 m glucose solution in water. $K_b = 0.512$ K kg mol^{-1}.

63. [M] Why is a semipermeable membrane essential for osmotic pressure measurements?

64. [H] A 0.05 M solution has osmotic pressure 1.23 atm at 300 K. Calculate its van't Hoff factor using $R = 0.0821$ L atm mol^{-1} K^{-1}.

65. [M] State one limitation of using boiling-point elevation to determine molar mass.

66. [H] A solute dissociates into three ions and has degree of dissociation 0.60. Calculate its van't Hoff factor.

67. [M] What does a van't Hoff factor equal to one indicate in an ideal dilute solution?

68. [H] A compound associates to form trimers. If the degree of association is α, derive the expression for i.

69. [M] Why is the freezing-point depression method unsuitable for some thermally unstable solutes?

70. [H] Explain how experimental molar mass differing from formula molar mass can reveal association or dissociation.

71. [M] A solution contains a volatile solute and a volatile solvent. How does each component contribute to total vapour pressure?

72. [H] A 0.20 m solution of $CaCl_2$ is assumed to dissociate completely. Calculate its boiling-point elevation in water using $K_b = 0.512$ K kg mol^{-1}.

UNIT 2: ELECTROCHEMISTRY

Questions 73–144

73. [M] Define oxidation and reduction in terms of electron transfer.

74. [M] Distinguish between a galvanic cell and an electrolytic cell.

75. [M] What is the EMF of an electrochemical cell?

76. [M] Define standard electrode potential and state the standard conditions.

77. [H] Explain why the standard hydrogen electrode is assigned a potential of zero volts.

78. [M] Write the cell notation for a Daniell cell.

79. [H] Derive the relation between standard cell potential and standard Gibbs energy change.

80. [M] State the Nernst equation for a general cell reaction.

81. [H] Write the Nernst equation for the cell $Zn(s)|Zn^{2+}(aq)||Cu^{2+}(aq)|Cu(s)$ at 298 K.

82. [M] What is the relationship between cell potential and spontaneity of a cell reaction?

83. [H] A cell reaction has n = 2 and E°_{cell} = 1.10 V. Calculate ΔG° using F = 96500 C mol^{-1}.

84. [M] Define the equilibrium constant of a cell reaction.

85. [H] Derive the relation between E°_{cell} and the equilibrium constant at 298 K.

86. [M] What is the function of a salt bridge in a galvanic cell?

87. [H] Explain why a galvanic cell stops producing current when its cell reaction reaches equilibrium.

88. [M] Define specific conductivity and state its SI unit.

89. [M] Define molar conductivity and state its usual unit.

90. [H] Explain why conductivity decreases on dilution while molar conductivity generally increases.

91. [M] How does molar conductivity vary with concentration for strong and weak electrolytes?

92. [H] Explain why the increase in molar conductivity on dilution is much greater for weak electrolytes.

93. [M] State Kohlrausch's law of independent migration of ions.

94. [H] Show how Kohlrausch's law can be used to determine the limiting molar conductivity of a weak electrolyte.

95. [M] How can the degree of dissociation of a weak electrolyte be obtained from conductivity data?

96. [H] Explain how the dissociation constant of a weak electrolyte can be calculated using conductivity measurements.

97. [M] State Faraday's first law of electrolysis.

98. [M] State Faraday's second law of electrolysis.

99. [H] Calculate the charge required to deposit one mole of silver from Ag^+ ions.

100. [H] How much copper is deposited by passing 2.0 A current for 30 minutes through $CuSO_4$ solution? Use F = 96500 C mol^{-1} and Cu = 63.5 g mol^{-1}.

101. [M] What is the role of the electrolyte in an electrolytic cell?

102. [H] Compare the products of electrolysis of aqueous NaCl and molten NaCl.

103. [M] Describe the construction and working principle of a dry cell.

104. [M] Describe the main components of a lead accumulator.

105. [H] Write the electrode reactions occurring during discharge of a lead accumulator.

106. [M] What is a fuel cell? Give one advantage over a conventional cell.

107. [H] Write the anode, cathode and overall reactions for a hydrogen–oxygen fuel cell in alkaline medium.

108. [M] Why is corrosion considered an electrochemical process?

109. [H] Explain the electrochemical mechanism of rusting of iron.

110. [M] What is the difference between primary and secondary batteries?

111. [H] A cell has $E°_{cell}$ = 0.46 V. Explain how the sign of $\Delta G°$ is determined and what it implies.

112. [M] Define the electrochemical series and mention two applications.

113. [H] Use standard electrode potentials to explain how to predict whether one metal can displace another from its salt solution.

114. [M] What is the standard electrode potential of an electrode measured against?

115. [H] Explain why E_{cell} changes as the concentrations of ions change during cell operation.

116. [M] Define limiting molar conductivity.

117. [H] Explain the difference between ionic mobility and ionic conductivity.

118. [M] Why does temperature generally increase the conductivity of an electrolytic solution?

119. [H] A conductivity cell has resistance 200 Ω and cell constant 1.25 cm^{-1}. Calculate conductivity in S cm^{-1}.

120. [H] A 0.01 M electrolyte solution has conductivity 1.5 × 10^{-3} S cm^{-1}. Calculate its molar conductivity in S cm^2 mol^{-1}.

121. [M] Explain why electrolytic conduction differs from metallic conduction.

122. [H] A reaction has E°cell = 0.00 V. What can be inferred about ΔG° and the equilibrium constant at standard conditions?

UNIT 2 — Additional Practice

123. [M] What is the direction of electron flow in an operating Daniell cell?

124. [H] Given E°(Cu^{2+}/Cu) = +0.34 V and E°(Zn^{2+}/Zn) = −0.76 V, calculate E°cell for a Daniell cell.

125. [M] Why is the anode negative in a galvanic cell but positive in an electrolytic cell?

126. [H] For a cell reaction involving two electrons, write ΔG° in terms of E°cell and Faraday constant.

127. [M] How does a salt bridge maintain electrical neutrality in the two half-cells?

128. [H] Use the Nernst equation to predict how Ecell changes when the concentration of the oxidised species increases.

129. [M] What is the reaction quotient Q in the Nernst equation?

130. [H] For a cell reaction with Q = 100, n = 2 and E°cell = 0.50 V at 298 K, calculate Ecell.

131. [M] Why is molar conductivity reported per mole of electrolyte rather than per unit volume?

132. [H] Explain how ionic inter-attraction affects molar conductivity at finite concentration.

133. [M] What is cell constant and how is it determined?

134. [H] A conductivity cell has conductance 0.005 S and cell constant 0.8 cm^{-1}. Calculate conductivity.

135. [M] Why does a weak electrolyte ionise more on dilution?

136. [H] Use Kohlrausch's law to express $\Lambda^\circ_m(CH_3COOH)$ using limiting ionic conductivities of suitable strong electrolytes.

137. [M] What is electrochemical equivalent?

138. [H] Calculate the time needed to deposit 1.08 g Ag using a 1.0 A current. Ag = 108 g mol^{-1}; F = 96500 C mol^{-1}.

139. [M] What is overvoltage in electrolysis?

140. [H] Explain why electrolysis of aqueous sodium sulfate can produce gases rather than sodium metal.

141. [M] Why is lead dioxide used as the positive active material in a lead accumulator?

142. [H] Compare the energy conversion in a hydrogen fuel cell and an electrolytic cell.

143. [M] Why is the standard electrode potential an intensive property?

144. [H] A concentration cell is formed using the same metal electrode in 0.01 M and 1.0 M solutions of its ion at 298 K. Explain the origin and direction of the cell potential.

UNIT 3: CHEMICAL KINETICS

Questions 145–216

145. [M] Define the rate of a chemical reaction.

146. [M] Distinguish between average rate and instantaneous rate.

147. [H] Explain how the rate of disappearance of a reactant is related to the rate of formation of a product using stoichiometric coefficients.

148. [M] Define rate law and rate constant.

149. [M] What is the order of a reaction?

150. [M] Define molecularity and distinguish it from order.

151. [H] Why can the order of a reaction be zero or fractional, whereas molecularity cannot?

152. [M] What is a pseudo-first-order reaction? Give an example.

153. [H] Explain why hydrolysis of an ester in excess water can behave as a pseudo-first-order reaction.

154. [M] Write the integrated rate equation for a zero-order reaction.

155. [M] Write the integrated rate equation for a first-order reaction.

156. [M] Write the half-life expression for a zero-order reaction.

157. [M] Write the half-life expression for a first-order reaction.

158. [H] Show that the half-life of a first-order reaction is independent of the initial concentration.

159. [H] A first-order reaction is 75% complete in 40 minutes. Calculate its half-life.

160. [H] For a first-order reaction, 20% of the reactant remains after 30 minutes. Calculate the rate constant.

161. [H] A zero-order reaction has $k = 2.0 \times 10^{-3}$ mol L^{-1} s^{-1} and initial concentration 0.10 mol L^{-1}. Calculate the time required for the concentration to fall to 0.04 mol L^{-1}.

162. [M] What is the unit of the rate constant for a zero-order reaction?

163. [M] What is the unit of the rate constant for a first-order reaction?

164. [H] Explain how a plot of concentration versus time can be used to identify a zero-order reaction.

165. [H] Explain how a plot of ln[A] versus time can be used to identify a first-order reaction.

166. [M] State the Arrhenius equation.

167. [M] Define activation energy.

168. [H] Explain how a catalyst changes the rate of a reaction without changing the equilibrium constant.

169. [M] What is collision theory?

170. [H] According to collision theory, why do not all molecular collisions lead to products?

171. [M] Define effective collision.

172. [H] Explain the role of proper orientation in collision theory.

173. [M] How does increasing concentration generally affect reaction rate?

174. [M] How does increasing temperature generally affect reaction rate?

175. [H] Explain why a small rise in temperature can cause a large increase in rate constant.

176. [H] A reaction rate constant doubles when temperature increases from 300 K to 310 K. Explain how activation energy can be estimated from this information.

177. [M] What is the frequency factor in the Arrhenius equation?

178. [H] Derive the two-temperature form of the Arrhenius equation.

179. [M] What is the significance of a plot of ln k against 1/T?

180. [H] How can activation energy be calculated from the slope of an Arrhenius plot?

181. [M] Distinguish between a slow reaction and a reaction with high activation energy.

182. [H] Explain why the rate law must generally be determined experimentally.

183. [M] For a reaction with rate = $k[A]^2[B]$, state the overall order.

184. [H] For rate = $k[A]^2[B]$, predict the factor by which the rate changes if [A] is doubled and [B] is tripled.

185. [M] What is the rate-determining step in a multistep reaction?

186. [H] Explain why the molecularity of an elementary step can be used to write its rate law, but the overall reaction equation usually cannot.

187. [M] What is the difference between a reaction mechanism and the overall reaction?

188. [H] A first-order reaction has $k = 6.93 \times 10^{-3}\ \text{min}^{-1}$. Calculate its half-life.

189. [M] What fraction of a first-order reactant remains after two half-lives?

190. [H] A zero-order reaction takes 20 minutes for concentration to decrease from 0.10 M to 0.06 M. Calculate its rate constant.

191. [M] How does a catalyst affect the activation energy of the forward and reverse reactions?

192. [H] Explain why a catalyst does not alter the position of chemical equilibrium.

193. [M] What is the difference between thermal and photochemical reactions in terms of energy input?

194. [H] Explain the limitations of the elementary collision theory model for complex reactions.

UNIT 3 — Additional Practice

195. [M] For rate = $k[A][B]^2$, state the order with respect to each reactant and the overall order.

196. [H] If rate = $k[A]^2[B]$, what happens to the rate when [A] is halved and [B] is doubled?

197. [M] Why is the rate constant independent of reactant concentration at a fixed temperature?

198. [H] A first-order reaction is 90% complete in 50 min. Calculate k.

199. [M] How much of a first-order reactant remains after three half-lives?

200. [H] A zero-order reaction has initial concentration 0.80 M and k = 0.02 M min^{-1}. Find its half-life.

201. [M] What does a negative slope in a first-order integrated-rate plot represent?

202. [H] Explain how half-life data at different initial concentrations can distinguish zero-order and first-order reactions.

203. [M] Why does a catalyst provide an alternative reaction pathway?

204. [H] Using the Arrhenius equation, explain how a plot of log k against 1/T differs in slope from a plot of ln k against 1/T.

205. [M] What is the threshold energy in collision theory?

206. [H] Explain why a reaction with a large activation energy is especially sensitive to temperature.

207. [M] Can a reaction have a negative order with respect to a reactant? Explain briefly.

208. [H] Explain how product inhibition or adsorption can lead to unusual experimentally observed reaction orders.

209. [M] What is the difference between rate and rate constant?

210. [H] For a reaction that is first order in A and zero order in B, write the rate law and describe the effect of doubling B.

211. [M] Why does the unit of the rate constant depend on reaction order?

212. [H] A first-order reaction has $k = 0.0231\ \text{min}^{-1}$. Calculate the time for 90% completion.

213. [M] What is meant by the effective collision energy?

214. [H] Explain the difference between activation energy and enthalpy change of a reaction.

215. [M] Why can catalysts be selective?

216. [H] For a first-order reaction, the time required for 99% completion is how many times its half-life? Derive the result.

UNIT 4: d- AND f-BLOCK ELEMENTS

Questions 217–288

217. [M] Define transition elements in terms of partially filled d orbitals.

218. [M] Write the general electronic configuration of d-block elements.

219. [H] Why are Zn, Cd and Hg generally not considered transition elements under the strict definition?

220. [M] Why do transition metals show variable oxidation states?

221. [H] Explain why the +2 oxidation state is common among first-row transition metals.

222. [M] Why are many transition-metal ions coloured?

223. [H] Explain d–d transitions and their role in the colour of transition-metal complexes.

224. [M] Why are d^0 and d^{10} ions often colourless?

225. [H] Explain why the colour of a transition-metal ion can change with ligand or oxidation state.

226. [M] Why do transition metals often act as catalysts?

227. [H] Explain how variable oxidation states and surface adsorption contribute to catalytic activity.

228. [M] What is the origin of paramagnetism in transition-metal ions?

229. [H] Calculate the spin-only magnetic moment for an ion containing three unpaired electrons.

230. [M] Why do transition metals form alloys readily?

231. [M] What are interstitial compounds? Give two characteristic properties.

232. [H] Explain why interstitial compounds are often hard and have high melting points.

233. [M] Describe the general trend in atomic radii across the first transition series.

234. [H] Why is the variation in ionisation enthalpy across the first transition series irregular?

235. [M] What is lanthanoid contraction?

236. [H] Explain the cause of lanthanoid contraction in terms of shielding by 4f electrons.

237. [M] State two consequences of lanthanoid contraction.

238. [H] Explain the similarity in radii of zirconium and hafnium.

239. [M] Write the general electronic configuration of lanthanoids.

240. [M] Why is +3 the most common oxidation state of lanthanoids?

241. [H] Explain why some lanthanoids also exhibit +2 or +4 oxidation states.

242. [M] Write the general electronic configuration of actinoids.

243. [H] Why do actinoids show a wider range of oxidation states than lanthanoids?

244. [M] Give two differences between lanthanoids and actinoids.

245. [H] Explain why most actinoids are radioactive.

246. [M] What is the colour of potassium dichromate solution and which chromium oxidation state is present?

247. [M] What is the colour of permanganate ion in aqueous solution?

248. [H] Write the balanced ionic equation for the reduction of dichromate ion to Cr^{3+} in acidic medium.

249. [H] Write the balanced ionic equation for the reduction of permanganate ion to Mn^{2+} in acidic medium.

250. [M] Describe the oxidising action of acidified potassium permanganate.

251. [H] Explain why the products of permanganate reduction depend on the pH of the medium.

252. [M] State the oxidation state of chromium in $K_2Cr_2O_7$.

253. [M] State the oxidation state of manganese in $KMnO_4$.

254. [H] Calculate the oxidation state of manganese in MnO_4^{2-} and compare it with MnO_4^-.

255. [M] Why is Cr^{2+} a reducing agent while Mn^{3+} can act as an oxidising agent?

256. [H] Use electronic configurations to explain the relative stability of Cr^{3+} and Mn^{2+}.

257. [M] Why do transition metals have high enthalpies of atomisation?

258. [H] Explain the trend in melting points across the first-row transition series and why it is not perfectly regular.

259. [M] What is disproportionation? Give a transition-metal example.

260. [H] Explain why Cu^+ tends to disproportionate in aqueous solution under standard conditions.

261. [M] Why are transition-metal compounds frequently used as pigments?

262. [H] Compare the magnetic behaviour expected for high-spin d^5 and low-spin d^6 complexes.

263. [M] What is the difference between lanthanoid and actinoid contraction?

264. [H] Explain why separation of lanthanoids is difficult and how contraction helps distinguish their properties.

265. [M] Why is the +6 oxidation state accessible to chromium in chromate and dichromate ions?

266. [H] Explain the interconversion of chromate and dichromate ions with change in pH.

UNIT 4 — Additional Practice

267. [M] Write the electronic configuration of Cr and explain its exceptional configuration.

268. [H] Explain the exceptional electronic configurations of Cr and Cu in terms of subshell stability.

269. [M] Why is the third ionisation enthalpy of manganese relatively high after formation of Mn^{2+}?

270. [H] Compare the stability of Fe^{2+} and Fe^{3+} using electronic configurations.

271. [M] What is the common oxidation state of scandium and why?

272. [H] Explain why the highest oxidation states are more common for early transition elements than for later ones.

273. [M] Why do transition-metal ions form complex ions readily?

274. [H] Explain the relationship between d-electron configuration and the magnetic behaviour of transition-metal ions.

275. [M] Why is Cu^{2+} coloured but Zn^{2+} colourless?

276. [H] Explain why d–d transition energies are affected by ligand identity.

277. [M] What is the effect of lanthanoid contraction on basicity of lanthanoid hydroxides?

278. [H] Explain the trend in basic strength of lanthanoid hydroxides across the series.

279. [M] Why are actinoids more prone to complex formation than lanthanoids?

280. [H] Explain why actinoid chemistry is more complicated due to 5f, 6d and 7s orbital energies.

281. [M] What is the role of sulfuric acid in preparing potassium dichromate from chromite ore?

282. [H] Balance the redox reaction between MnO_4^- and Fe^{2+} in acidic medium.

283. [M] Why is permanganate a stronger oxidising agent in acidic medium than in neutral medium?

284. [H] Explain the use of $K_2Cr_2O_7$ as a primary standard in redox titrations.

285. [M] Why are transition metals usually good conductors of heat and electricity?

286. [H] Explain why the enthalpies of atomisation of Zn and Cd are comparatively lower than those of many transition metals.

287. [M] What is the difference between a transition metal and an inner-transition metal?

288. [H] Explain why the standard electrode potential of a transition-metal couple cannot be predicted from electronic configuration alone.

UNIT 5: COORDINATION COMPOUNDS

Questions 289–358

289. [M] Define a coordination compound and give one example.

290. [M] What is a ligand? Classify ligands on the basis of denticity.

291. [M] Define coordination number with an example.

292. [H] Determine the oxidation state of the central metal in $[Co(NH_3)_5Cl]Cl_2$.

293. [M] Distinguish between a double salt and a coordination compound.

294. [M] State Werner's primary and secondary valencies.

295. [H] Explain how Werner's theory accounts for the structures of $CoCl_3 \cdot 6NH_3$ and $CoCl_3 \cdot 5NH_3$.

296. [M] Write the IUPAC name of $[Co(NH_3)_6]Cl_3$.

297. [M] Write the IUPAC name of $K_4[Fe(CN)_6]$.

298. [H] Write the formula of pentaamminechloridocobalt(III) chloride.

299. [M] What is a chelating ligand? Give an example.

300. [H] Explain why chelate complexes are often more stable than analogous complexes with monodentate ligands.

301. [M] What is the difference between homoleptic and heteroleptic complexes?

302. [M] Explain the bonding in coordination compounds according to valence bond theory.

303. [H] Predict the hybridisation and geometry of $[Ni(CN)_4]^{2-}$ using VBT.

304. [H] Predict the hybridisation and magnetic behaviour of $[NiCl_4]^{2-}$ using VBT.

305. [M] What is crystal field splitting?

306. [M] Distinguish between strong-field and weak-field ligands.

307. [H] Explain why $[Fe(CN)_6]^{4-}$ is low spin while $[FeF_6]^{4-}$ is generally high spin.

308. [M] Draw or describe the splitting of d orbitals in an octahedral field.

309. [M] Draw or describe the splitting of d orbitals in a tetrahedral field.

310. [H] Why is tetrahedral splitting smaller than octahedral splitting for the same metal and ligand?

311. [M] What is crystal field stabilisation energy?

312. [H] Calculate the CFSE for an octahedral d^3 ion, ignoring pairing energy.

313. [M] Why are many coordination compounds coloured?

314. [H] Explain how ligand-field splitting determines the wavelength of light absorbed by a complex.

315. [M] How can magnetic measurements help distinguish high-spin and low-spin complexes?

316. [H] Calculate the spin-only magnetic moment of a complex with four unpaired electrons.

317. [M] What is geometrical isomerism in coordination compounds?

318. [M] What is optical isomerism in coordination compounds?

319. [H] Explain cis–trans isomerism in square-planar complexes of type $[MA_2B_2]$.

320. [H] How many geometrical isomers are possible for an octahedral complex of type $[MA_4B_2]$?

321. [M] Give an example of a coordination compound showing linkage isomerism.

322. [M] What is ionisation isomerism?

323. [H] Distinguish between linkage, ionisation and hydrate isomerism with examples.

324. [M] Explain the role of coordination compounds in qualitative analysis.

325. [M] How is complex formation used in the extraction of gold or silver?

326. [M] State the role of haemoglobin as a coordination compound.

327. [H] Explain the role of chlorophyll and vitamin B_{12} in biological systems in terms of their central metal ions.

328. [M] Why does EDTA form stable complexes with many metal ions?

329. [H] Compare inner-orbital and outer-orbital octahedral complexes in VBT.

330. [M] What is the spectrochemical series?

331. [H] Explain why pairing may occur in a strong ligand field but not in a weak ligand field.

332. [M] Determine the oxidation state and coordination number of Fe in $[Fe(CN)_6]^{3-}$.

333. [H] Predict the number of unpaired electrons in $[Mn(CN)_6]^{3-}$ using crystal field theory.

334. [M] What is the effective atomic number rule? State one limitation.

335. [H] Explain why the EAN rule does not reliably predict the stability of every coordination compound.

336. [M] Distinguish between a monodentate and a bidentate ligand.

337. [H] Explain why $[Pt(NH_3)_2Cl_2]$ has cis and trans forms and state the relevance of the cis form.

UNIT 5 — Additional Practice

338. [M] Find the oxidation state of Co and coordination number in $[Co(en)_2Cl_2]^+$.

339. [H] Write the IUPAC name of $[Cr(H_2O)_4Cl_2]Cl$.

340. [M] What is the denticity of ethane-1,2-diamine (en)?

341. [H] Predict the geometry and hybridisation of $[Fe(CN)_6]^{3-}$ using the strong-field assumption.

342. [M] Why is $[Zn(NH_3)_4]^{2+}$ generally diamagnetic?

343. [H] Compare the magnetic moments expected for high-spin and low-spin d^6 octahedral complexes.

344. [M] What is the difference between crystal-field splitting energy and pairing energy?

345. [H] For an octahedral d^4 ion, compare high-spin and low-spin electron arrangements.

346. [M] Give an example of a complex that exhibits optical isomerism.

347. [H] Explain why cis-$[M(en)_2Cl_2]^+$ can be optically active while its trans form may not be.

348. [M] What is coordination sphere?

349. [H] Determine the number of ions produced when $[Co(NH_3)_5Cl]Cl_2$ dissolves completely in water, ignoring complex dissociation.

350. [M] Why do coordination compounds often have specific geometries?

351. [H] Explain the limitations of VBT in explaining colour and spectra of coordination compounds.

352. [M] What is the chelate effect?

353. [H] Explain how coordination compounds can mask interfering ions in qualitative analysis.

354. [M] Why is $[Ag(NH_3)_2]^+$ important in qualitative analysis?

355. [H] Explain how complex formation can alter the solubility of a sparingly soluble salt.

356. [M] Write the formula of tetraamminecopper(II) sulfate.

357. [H] Distinguish between fac and mer isomers of an octahedral $[MA_3B_3]$ complex.

358. [H] For an octahedral d^5 metal ion, compare the number of unpaired electrons in high-spin and low-spin cases.

UNIT 6: HALOALKANES AND HALOARENES

Questions 359–428

359. [M] Define haloalkanes and haloarenes, giving one example of each.

360. [M] Write the IUPAC name of $CH_3CHBrCH_2CH_3$.

361. [M] Explain why the carbon–halogen bond is polar.

362. [H] Compare the C–F, C–Cl, C–Br and C–I bond strengths and relate them to reactivity.

363. [M] What is nucleophilic substitution?

364. [M] Distinguish between SN1 and SN2 reactions.

365. [H] Explain why tertiary haloalkanes generally favour SN1 while primary haloalkanes generally favour SN2 under suitable conditions.

366. [M] What is racemisation in a nucleophilic substitution reaction?

367. [H] Explain the stereochemical outcome of an SN2 reaction at a chiral carbon.

368. [M] What is optical activity?

369. [H] Why can an SN1 reaction at a chiral centre lead to racemisation?

370. [M] Arrange primary, secondary and tertiary haloalkanes in order of SN1 reactivity, with justification.

371. [H] Predict the major product when 2-bromobutane reacts with aqueous KOH and explain the reaction type.

372. [M] What is dehydrohalogenation?

373. [H] Compare substitution and elimination products when a haloalkane reacts with aqueous KOH versus alcoholic KOH.

374. [M] Why are haloarenes less reactive than haloalkanes towards nucleophilic substitution?

375. [H] Explain the partial double-bond character of the C–Cl bond in chlorobenzene using resonance.

376. [M] Why does chlorobenzene direct electrophilic substitution to ortho and para positions?

377. [H] Explain why halogens are deactivating but ortho/para-directing in electrophilic aromatic substitution.

378. [M] Write the products of nitration of chlorobenzene.

379. [M] What is the Wurtz reaction?

380. [M] What is the Fittig reaction?

381. [M] What is the Wurtz–Fittig reaction?

382. [H] Predict the products when bromobenzene reacts with sodium in dry ether in the presence of methyl bromide.

383. [M] How can a haloalkane be converted into an alcohol?

384. [H] Suggest a route to convert ethanol into bromoethane and state a suitable reagent.

385. [M] What is the difference between allylic and vinylic halides?

386. [H] Explain why vinyl chloride is less reactive than ethyl chloride towards nucleophilic substitution.

387. [M] State one use and one environmental concern associated with dichloromethane.

388. [M] State one use and one health concern associated with chloroform.

389. [M] Why is carbon tetrachloride environmentally problematic?

390. [M] What is iodoform and what is one historical use of it?

391. [M] What are freons and why have many been phased out?

392. [M] Why is DDT considered an environmental pollutant?

393. [H] Compare the environmental persistence and bioaccumulation concerns of DDT and many modern pesticides.

394. [M] How does the nature of the halogen affect the boiling point of haloalkanes?

395. [H] Explain why branching generally lowers the boiling point of isomeric haloalkanes.

396. [M] Why are haloalkanes only slightly soluble in water?

397. [H] Explain why haloalkanes can be soluble in organic solvents despite having a polar C–X bond.

398. [M] What is a Grignard reagent?

399. [H] Why must Grignard reagents be prepared and used under anhydrous conditions?

400. [M] Write the reaction of bromoethane with magnesium in dry ether.

401. [H] Predict the product when a Grignard reagent reacts with carbon dioxide followed by acidic hydrolysis.

402. [M] How can the presence of halogen in an organic compound be tested after suitable preparation of the sample?

403. [H] Explain why direct treatment of an organic halide with aqueous silver nitrate may not give a reliable result in every case.

404. [M] Distinguish between nucleophiles and bases.

405. [H] Explain how solvent polarity and solvent proticity can affect SN1 and SN2 reactions.

406. [M] What is the role of a leaving group in nucleophilic substitution?

407. [H] Arrange typical halide leaving groups in order of leaving-group ability and justify the trend.

UNIT 6 — Additional Practice

408. [M] Write the IUPAC name of $CH_3CH_2CHClCH_3$.

409. [H] Predict the major product of 2-bromopropane with alcoholic KOH and explain the mechanism.

410. [M] Why does tertiary butyl chloride react rapidly by SN1 in a polar protic solvent?

411. [H] Explain the energy profile of SN1 and SN2 mechanisms.

412. [M] What is inversion of configuration?

413. [H] A chiral haloalkane undergoes substitution with a strong nucleophile in a polar aprotic solvent. Predict the likely stereochemical outcome.

414. [M] Why is chlorobenzene resistant to hydrolysis under ordinary conditions?

415. [H] Explain how electron-withdrawing groups at ortho and para positions facilitate nucleophilic substitution in haloarenes.

416. [M] What is the product of hydrolysis of bromoethane with aqueous KOH?

417. [H] Give a sequence to convert propene into 1-bromopropane, noting the required conditions.

418. [M] What is the role of dry ether in the Wurtz reaction?

419. [H] Why can Wurtz reactions of two different haloalkanes give a mixture of alkanes?

420. [M] Why are CFCs harmful to stratospheric ozone?

421. [H] Explain the chain mechanism by which chlorine radicals destroy ozone, at a conceptual level.

422. [M] What is the difference between a geminal and a vicinal dihalide?

423. [H] Compare the reactivity of allyl chloride, ethyl chloride and vinyl chloride toward nucleophilic substitution.

424. [M] Why are organohalogen compounds often denser than corresponding hydrocarbons?

425. [H] Explain how a chiral centre can be formed during a substitution reaction and how its configuration is assigned.

426. [M] What is the major organic product when bromoethane reacts with potassium cyanide?

427. [H] Compare the products obtained when bromoethane reacts with KCN and AgCN.

428. [H] Explain why the rate of an SN2 reaction depends on both haloalkane and nucleophile concentrations.

UNIT 7: ALCOHOLS, PHENOLS AND ETHERS

Questions 429–500

429. [M] Define alcohols, phenols and ethers with one example of each.

430. [M] Write the IUPAC name of $CH_3CH_2CH(OH)CH_3$.

431. [M] Distinguish between primary, secondary and tertiary alcohols.

432. [M] Why do alcohols have higher boiling points than comparable alkanes?

433. [H] Explain how hydrogen bonding affects the boiling points and water solubilities of alcohols.

434. [M] Describe one method for preparing ethanol from ethene.

435. [M] How can a haloalkane be converted into an alcohol?

436. [H] Explain the preparation of alcohols by reduction of aldehydes and ketones.

437. [M] What is the oxidation product of a primary alcohol under controlled oxidation?

438. [H] Compare the oxidation products of primary, secondary and tertiary alcohols under common conditions.

439. [M] What is the Lucas test? What does it help distinguish?

440. [H] Explain the relative rates of reaction of primary, secondary and tertiary alcohols with Lucas reagent.

441. [M] What happens when ethanol reacts with sodium metal?

442. [M] What is dehydration of an alcohol?

443. [H] Describe the mechanism of acid-catalysed dehydration of ethanol to ethene.

444. [M] Why is concentrated sulfuric acid used in the dehydration of alcohols?

445. [H] Explain why tertiary alcohols generally dehydrate more readily than primary alcohols.

446. [M] What is the product when ethanol is heated with excess concentrated sulfuric acid at about 413 K?

447. [H] Compare the products formed by heating ethanol with concentrated sulfuric acid at about 413 K and 443 K.

448. [M] Define phenol and write its formula.

449. [M] Why is phenol more acidic than ethanol?

450. [H] Explain the greater stability of the phenoxide ion compared with the ethoxide ion.

451. [M] How does an electron-withdrawing group affect the acidity of phenol?

452. [H] Compare the acidity of phenol, p-nitrophenol and p-cresol with justification.

453. [M] Describe the preparation of phenol from chlorobenzene.

454. [M] What happens when phenol reacts with bromine water?

455. [H] Explain why phenol undergoes electrophilic substitution more readily than benzene.

456. [M] What is the Kolbe–Schmitt reaction?

457. [M] What is the Reimer–Tiemann reaction?

458. [H] Write the main product of the Reimer–Tiemann reaction of phenol and explain the role of the reagent.

459. [M] What happens when phenol reacts with dilute nitric acid?

460. [H] Compare the products of nitration of phenol under dilute and concentrated nitric acid conditions.

461. [M] How can phenol be distinguished from ethanol using a simple chemical test?

462. [M] Define ethers and give the general structure of an ether.

463. [M] Write the IUPAC name of $CH_3OCH_2CH_3$.

464. [M] Describe the Williamson ether synthesis.

465. [H] Why is a primary alkyl halide generally preferred in Williamson ether synthesis?

466. [H] Predict the major product when sodium ethoxide reacts with 2-bromopropane and explain the competing reaction.

467. [M] What happens when an ether is heated with excess HI?

468. [H] Explain the cleavage of an unsymmetrical ether by HI and how the products depend on the alkyl groups.

469. [M] Why do ethers have lower boiling points than isomeric alcohols?

470. [M] State one use of diethyl ether.

471. [H] Explain why ethers can form explosive peroxides on prolonged exposure to air and light.

472. [M] What is the effect of methanol on human health?

473. [M] Why is ethanol used as a solvent in many products?

474. [H] Compare methanol and ethanol in terms of preparation, uses and toxicity.

475. [M] What is esterification of an alcohol?

476. [H] Write the reaction of ethanol with ethanoic acid and state the role of the acid catalyst.

477. [M] How does the solubility of alcohols in water change with increasing carbon-chain length?

478. [H] Explain why phenol is only moderately soluble in water despite its ability to form hydrogen bonds.

479. [M] How can primary, secondary and tertiary alcohols be distinguished by oxidation behaviour?

UNIT 7 — Additional Practice

480. [M] Write the IUPAC name of $(CH_3)_3COH$.

481. [H] Predict the major product when 2-methylpropan-2-ol is dehydrated under acidic conditions.

482. [M] Why does phenol react with sodium hydroxide but ethanol does not react appreciably?

483. [H] Explain how resonance and inductive effects together influence the acidity of substituted phenols.

484. [M] What is the product when phenol reacts with bromine in a non-polar solvent under controlled conditions?

485. [H] Explain the difference between bromination of phenol in bromine water and in a non-polar solvent.

486. [M] How is anisole prepared by Williamson ether synthesis?

487. [H] Predict the products of anisole cleavage with excess HI and explain which bond is cleaved first.

488. [M] Why is the C–O bond in phenol shorter than a typical C–O single bond in an alcohol?

489. [H] Explain why alcohol dehydration generally follows the order tertiary > secondary > primary under acidic conditions.

490. [M] What is the product of oxidation of propan-2-ol?

491. [H] How can 1-propanol be distinguished from 2-propanol by oxidation products?

492. [M] What is the role of acid in esterification?

493. [H] Give a reaction sequence to convert phenol into salicylic acid.

494. [M] What is the product of phenol with neutral ferric chloride solution?

495. [H] Explain the limitations of using the ferric chloride test as the only evidence for phenol.

496. [M] Why are lower alcohols miscible with water?

497. [H] Explain the effect of branching on the boiling points of isomeric alcohols.

498. [M] What is the major product when diethyl ether reacts with excess HI on heating?

499. [H] Explain why ethers are relatively unreactive toward bases but can be cleaved by strong hydrogen halides.

500. [H] Predict the products when phenol reacts with aqueous NaOH and then with CO_2 under pressure followed by acidification. Name the reaction.